

CV ASSIGNMENT 1 Date: _____QUESTION 01

The paper investigates whether modern generative models actually reproduce the rules of projective geometry when generating realistic images. Although previous studies suggest that models implicitly capture scene properties such as depth, normals and albedo, this paper shows that generated images can still contain systematic geometric errors. In particular, generated images may have inconsistent vanishing points and perspective, as well as incorrect object-shadow relationships. These errors matter because they reveal a limitation in the geometric understanding of current generative models and provide a way to evaluate generated images using geometry rather than only pixel level artifacts.

QUESTION 02

The main idea is to detect and analyze generated images using derived geometric cues

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instead of raw image pixels. The authors develop three classifiers based on object-shadow relationships, perspective fields and line segments. A pixel based ~~ResNet~~ ResNet-50 qualifier is first used to filter out images that are easy to distinguish using ordinary image signals, leaving difficult "unconfident" and misclassified examples. The geometry based classifiers can then identify generated images even when pixel level artifacts are insufficient. The paper also demonstrates that these geometric cues generalize to several unseen generators, showing that the geometric errors are systematic rather than isolated artifacts.

QUESTION 03

The method consists of four main stages:

① Data preparation and prequalifier:

Real images are collected from L.SUN for indoor scenes and BDD100K and Mapillary Vistas for outdoor scenes.

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Captions are generated for the real images and the same captions are used to generate images using Stable Diffusion XL, Kandinsky-v3, DeepFloyd IF and PixArt- α . A pixel based ResNet-50 prequalifier separates images into easy, unconfident and misclassified sets. The main analysis focuses on the difficult unconfident and misclassified sets.

② Object-Shadow classifier:

A pretrained object-shadow detector, SSISv2, extracts object and shadow masks. A ResNet-50 is then trained on these binary masks to determine whether the relationships between objects and their shadows are geometrically plausible.

③ Perspective Field classifier:

A pretrained perspective fields model converts an image into geometric vector fields describing spatial orientation relative to vanishing points and the horizon. A ResNet-50 analyzes these fields to detect perspective inconsistencies.

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④ Line Segment classifier:

DeepLSD detects structural line segments. These are represented as an unordered set and passed to a PointNet-like network, which analyzes their spatial arrangement and detect inconsistencies in perspective geometry.

The geometry classifiers are trained using Kandinsky-v3 data and achieve AUCs ranging from approximately 0.72 to 0.97 in the main evaluation. They also generalize to unseen generators such as SDXL, Pixart- α , DALL-E 3 and Adobe Firefly.

QUESTION 04

Datasets:

The paper uses 400000 indoor real images from LSUN and 125000 outdoor real images from BDD100K and Mapillary Vistas. It also uses a recent timestamp set containing 500 indoor and 500 outdoor real images. Generated images

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are produced using Stable Diffusion XL, Kandinsky-v3, DeepFloyd IF and PixArt- α using the same captions as the real images. Generalization is additionally tested on DALL-E 3 and Adobe Firefly.

Baselines / comparisons:

A ResNet-50 pixel based prequalifier is used to identify images that can be detected using ordinary image signals and to create difficult test sets. The paper also compares its geometry classifiers against basic statistical cues with logistic regression, ~~using~~ including line counts / lengths / orientations, object-shadow counts and perspective-field statistics.

QUESTION 05

Yes, the method shows strong cross-generator generalization. Classifiers trained on Kandinsky-v3 can detect geometric inconsistencies in images generated by SDXL, PixArt- α , DALL-E 3 and Adobe Firefly without retraining. However, DeepFloyd IF is a major exception,

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where the Kandinsky-trained classifiers perform close to chance in several settings. A classifier trained specifically on DeepFloyd can detect its images, but it generalizes less effectively to other generators. A potential limitation is that outdoor and more complex scenes can be harder because they contain more varied structures and visual clutter. Errors in pretrained geometry extraction tools could also affect performance. Finally, future generators could reduce these errors by explicitly modeling projective geometry or using geometry-aware loss functions.

QUESTION 06

The paper raises several questions. Can generative models be trained with explicit geometry-aware losses, such as shadow consistency and vanishing point alignment, and would this reduce the detected errors? Why does DeepFloyd behave differently from the other generators? Could the three geometric cues be combined into a

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single detector to improve robustness?

Finally, the same idea can be extended to video or 3D generation, where temporal and multi-view geometric could provide even stronger tests?